

Package ‘manMetaVAR’

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Title Multivariate Meta-Analysis of Vector Autoregressive Model
Coefficients

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coef.manmetavar.dtvvar *Parameter Estimates (FitDTVAR)*

Description

Parameter Estimates (FitDTVAR)
 Parameter Estimates (FitMetaVAR)
 Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
```

```

    theta = TRUE,
    converged = TRUE,
    grad_tol = 0.01,
    hess_tol = 1e-08,
    vanishing_theta = TRUE,
    theta_tol = 0.001,
    ...
)

## S3 method for class 'manmetavar.metavar'
coef(object, ...)

## S3 method for class 'manmetavar.mplus'
coef(object, median = TRUE, burnin = NULL, ...)
```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
alpha	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
psi	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
theta	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
vanishing_theta	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
theta_tol	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
...	additional arguments.
median	Logical. If <code>median = TRUE</code> , return median of the posterior. If <code>median = FALSE</code> , return mean of the posterior.
burnin	Integer indicating initial samples to discard. If <code>burnin = NULL</code> , do not discard anything.

Value

- Returns a list of vectors of parameter estimates.
- Returns a vector of estimated parameters.
- Returns a vector of estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

Compress	<i>Compress Replication</i>
----------	-----------------------------

Description

Compress Replication

Usage

Compress(taskid, repid, output_folder)

Arguments

- | | |
|---------------|-----------------------------------|
| taskid | Positive integer. Task ID. |
| repid | Positive integer. Replication ID. |
| output_folder | Character string. Output folder. |

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

 confint.manmetavar.metavar

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Description

Confidence Intervals for the Parameter Estimates (FitMetaVAR)

Confidence Intervals for the Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.metavar'
confint(object, parm = NULL, level = 0.95, lb = TRUE, ...)

## S3 method for class 'manmetavar.mplus'
confint(object, parm = NULL, level = 0.95, burnin = NULL, ...)
```

Arguments

object	Object of class manmetavar.mplus.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
lb	Logical. If TRUE, returns profile likelihood-based confidence intervals. If FALSE, returns Wald confidence intervals.
...	additional arguments.
burnin	Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.

Value

Returns a matrix of confidence intervals.

Returns a matrix of confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Dynamics	<i>Process Dynamics</i>
----------	-------------------------

Description

Process Dynamics

Usage

Dynamics(dynamics)

Arguments

dynamics 1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

See Also

Other Data Generation Functions: [GenData\(\)](#)

Examples

```
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)
```

FitDTVAR	<i>Fit the Model using the fitDTVARMxID Package</i>
----------	---

Description

The function fits the model using the [fitDTVARMxID::fitDTVARMxID](#) package.

Usage

FitDTVAR(data, ncores = NULL, seed = NULL)

Arguments

data R object. Output of the [GenData\(\)](#) function.
ncores Positive integer. Number of cores to use.
seed Integer. Random seed.

See Also

Other Model Fitting Functions: [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

FitMetaVAR

Multivariate Meta-Analysis using the metaVAR Package

Description

The function performs multivariate meta-analysis using the [metaVAR::metaVAR](#) package.

Usage

```
FitMetaVAR(fit, ncores = NULL)
```

Arguments

<code>fit</code>	R object. Output of the FitDTVAR() function.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
```

```
pooled <- FitMetaVAR(  
  fit = fit,  
  ncores = parallel::detectCores()  
)  
summary(pooled)  
print(pooled)  
coef(pooled)  
vcov(pooled)  
  
## End(Not run)
```

FitMLVAR*Fit the Model using the mlVAR Package*

Description

The function fits the model using the [mlVAR::mlVAR](#) package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

<code>data</code>	R object. Output of the GenData() function.
<code>ncores</code>	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:  
set.seed(42)  
data <- GenData(taskid = 1)  
fit <- FitMLVAR(data = data)  
print(fit)  
summary(fit)  
  
## End(Not run)
```


FitMplus

*Fit the Model using Mplus***Description**

The function fits the model using Mplus.

Usage

```
FitMplus(
  data,
  chains = 2L,
  iter = 120000L,
  fscores = NULL,
  plot = FALSE,
  default_priors = TRUE,
  wd = ".",
  mplus_bin = NULL,
  ncores = NULL
)
```

Arguments

data	R object. Output of the GenData() function.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.
wd	Character string. Working directory.
mplus_bin	Character string. Path to Mplus binary. If mplus_bin = NULL, the function will try to find the appropriate binary.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [MplusInput\(\)](#)

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMplus(data = data)
print(fit)
```

```
summary(fit)

## End(Not run)
```

GenData	<i>Simulate Data</i>
---------	----------------------

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

taskid Positive integer. Task ID.

See Also

Other Data Generation Functions: [Dynamics\(\)](#)

Examples

```
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)
```

MplusInput	<i>Create Mplus Input File</i>
------------	--------------------------------

Description

The function creates an Mplus input file.

Usage

```
MplusInput(
  dynamics,
  fn_data,
  fn_posterior,
  fn_factorscores,
  chains,
  iter,
  fscores,
  plot,
  default_priors,
  ncores = NULL
)
```

Arguments

dynamics	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.
fn_data	Character string. Filename for data file.
fn_posterior	Character string. Filename for posterior output.
fn_factorscores	Character string. Filename for factor scores output.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.
ncores	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: [FitDTVAR\(\)](#), [FitMLVAR\(\)](#), [FitMetaVAR\(\)](#), [FitMplus\(\)](#)

Examples

```
cat(
  MplusInput(
    dynamics = 1,
    fn_data = "data.dat",
    fn_posterior = "posterior.dat",
    fn_factorscores = "factorscores.dat",
    chains = 2L,
    iter = 60000L,
    fscores = 1000L,
    plot = TRUE,
    default_priors = FALSE,
    ncores = NULL
  )
)
```

```
)  
)
```

params	<i>Simulation Parameters</i>
--------	------------------------------

Description

Simulation Parameters

Usage

params

Format

A dataframe with 25 rows and 3 columns:

- taskid** Simulation Task ID.
- n** Sample size.
- time** Number of measurement occasions.
- dynamics** Process dynamics. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

<code>plot.manmetavar.data</code>	<i>Plot Method for an Object of Class manmetavar.data</i>
-----------------------------------	---

Description

- Plot Method for an Object of Class `manmetavar.data`
- Plot Method for an Object of Class `manmetavar.mplus`

Usage

```
## S3 method for class 'manmetavar.data'  
plot(x, ...)  
  
## S3 method for class 'manmetavar.mplus'  
plot(x, what = "posterior", parm = NULL, level = 0.95, burnin = NULL, ...)
```

Arguments

x	Object of class manmetavar.mplus.
...	additional arguments.
what	Character string. If what = "posterior", return posterior distribution plots. If what = "trace", return trace plots.
parm	a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered.
level	the confidence level required.
burnin	Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.data *Print Method for an Object of Class manmetavar.data*

Description

Print Method for an Object of Class manmetavar.data

Print Method (FitDTVAR)

Print Method (FitMetaVAR)

Print Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)

## S3 method for class 'manmetavar.dtvvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
```

```

    theta_tol = 0.001,
    digits = 4,
    ...
)

## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

<code>x</code>	Object of class <code>manmetavar.mlvar</code> .
<code>...</code>	additional arguments.
<code>means</code>	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
<code>alpha</code>	Numeric vector. Significance level α .
<code>beta</code>	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
<code>nu</code>	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
<code>psi</code>	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
<code>theta</code>	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
<code>converged</code>	Logical. Only include converged cases.
<code>grad_tol</code>	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
<code>hess_tol</code>	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
<code>vanishing_theta</code>	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
<code>theta_tol</code>	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .
<code>digits</code>	Integer indicating the number of decimal places to display.
<code>show</code>	Tables to show.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

Sim	<i>Simulation Replication</i>
-----	-------------------------------

Description

Simulation Replication

Usage

```
Sim(  
  taskid,  
  repid,  
  output_folder,  
  overwrite,  
  integrity,  
  seed,  
  data,  
  metavar,  
  mlvar,  
  mplus,  
  chains,  
  iter,  
  fscores,  
  plot,  
  default_priors  
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
seed	Integer. Random seed.
data	Logical. Simulate data.

metavar	Logical. If metavar = TRUE, fit the model using the metaVAR package.
mlvar	Logical. If mlvar = TRUE, fit the model using the mlVAR package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If plot = TRUE, add PLOT: TYPE = PLOT3; to Mplus input file.
default_priors	Logical. If default_priors = TRUE, use default priors.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDTVAR	<i>Simulation Replication - FitDTVAR</i>
-------------	--

Description

Simulation Replication - FitDTVAR

Usage

SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR

Simulation Replication - FitMetaVAR

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMLVAR

Simulation Replication - FitMLVAR

Description

Simulation Replication - FitMLVAR

Usage

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMplus

Simulation Replication - FitMplus

Description

Simulation Replication - FitMplus

Usage

```
SimFitMplus(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  chains,
  iter,
  fscores,
  plot,
  default_priors
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .
chains	Integer. Number of chains.
iter	Integer. Number of iterations.
fscores	Integer. Number of iterations for factor scores.
plot	Logical. If <code>plot = TRUE</code> , add <code>PLOT: TYPE = PLOT3;</code> to Mplus input file.
default_priors	Logical. If <code>default_priors = TRUE</code> , use default priors.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN	<i>Simulation File Name</i>
-------	-----------------------------

Description

Simulation File Name

Usage

SimFN(output_type, output_folder, suffix)

Arguments

- output_type Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
- output_folder Character string. Output folder.
- suffix Character string. Output of manCTMed::SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData	<i>Simulation Replication - GenData</i>
------------	---

Description

Simulation Replication - GenData

Usage

SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of <code>manCTMed:::SimSuffix()</code> .
overwrite	Logical. Overwrite existing output in <code>output_folder</code> .
integrity	Logical. If <code>integrity = TRUE</code> , check for the output file integrity when <code>overwrite = FALSE</code> .

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj

Simulation Project Name

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

Sum	<i>Summary</i>
-----	----------------

Description

Summary

Usage

```
Sum(taskid, reps, output_folder, overwrite, integrity, metavar, mlvar, mplus)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
metavar	Logical. If metavar = TRUE, fit the model using the metaVAR package.
mlvar	Logical. If mlvar = TRUE, fit the model using the mlVAR package.
mplus	Logical. If mplus = TRUE, fit the model using DSEM in Mplus.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARLB	<i>Summary: Likelihood-Based (FitMetaVAR)</i>
-----------------	---

Description

Summary: Likelihood-Based (FitMetaVAR)

Usage

```
SumFitMetaVARLB(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMetaVARNormal	<i>Summary: Normal Theory (FitMetaVAR)</i>
---------------------	--

Description

Summary: Normal Theory (FitMetaVAR)

Usage

```
SumFitMetaVARNormal(taskid, reps, output_folder, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMLVAR	<i>Summary (FitMLVAR)</i>
-------------	---------------------------

Description

Summary (FitMLVAR)

Usage

SumFitMLVAR(taskid, reps, output_folder, overwrite, integrity)

Arguments

- | | |
|---------------|---|
| taskid | Positive integer. Task ID. |
| reps | Positive integer. Number of replications. |
| output_folder | Character string. Output folder. |
| overwrite | Logical. Overwrite existing output in output_folder. |
| integrity | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SumFitMplus	<i>Summary (FitMplus)</i>
-------------	---------------------------

Description

Summary (FitMplus)

Usage

SumFitMplus(taskid, reps, output_folder, overwrite, integrity)

Arguments

- | | |
|---------------|---|
| taskid | Positive integer. Task ID. |
| reps | Positive integer. Number of replications. |
| output_folder | Character string. Output folder. |
| overwrite | Logical. Overwrite existing output in output_folder. |
| integrity | Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE. |

Details

This function is executed via the Sum function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar.data	<i>Summary Method for an Object of Class manmetavar.data</i>
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Description

- Summary Method for an Object of Class manmetavar.data
- Summary Method (FitDTVAR)
- Summary Method (FitMetaVAR)
- Summary Method (FitMLVAR)
- Summary Method (FitMplus)

Usage

```
## S3 method for class 'manmetavar.data'
summary(object, ...)

## S3 method for class 'manmetavar.dtvvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)

## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)

## S3 method for class 'manmetavar.mplus'
summary(object, alpha = 0.05, median = TRUE, digits = 4, burnin = NULL, ...)
```

Arguments

object	Object of class <code>manmetavar.mplus</code> .
...	additional arguments.
means	Logical. If <code>means = TRUE</code> , return means. Otherwise, the function returns raw estimates.
alpha	Numeric vector. Significance level α .
beta	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
nu	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.

psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
digits	Integer indicating the number of decimal places to display.
show	Tables to show.
median	Logical. If median = TRUE, return median of the posterior. If median = FALSE, return mean of the posterior.
burnin	Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Returns a matrix of estimates, standard errors, number of posterior samples, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

vcov.manmetavar.dtvvar *Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)*

Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Sampling Covariance Matrix of the Parameter Estimates (FitMplus)

Usage

```
## S3 method for class 'manmetavar.dtvvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)

## S3 method for class 'manmetavar.metavar'
vcov(object, ...)
```

```
## S3 method for class 'manmetavar.mplus'
vcov(object, burnin = NULL, ...)
```

Arguments

<code>object</code>	Object of class <code>manmetavar.mplus</code> .
<code>alpha</code>	Logical. If <code>alpha = TRUE</code> , include estimates of the alpha vector, if available. If <code>alpha = FALSE</code> , exclude estimates of the alpha vector.
<code>beta</code>	Logical. If <code>beta = TRUE</code> , include estimates of the beta matrix, if available. If <code>beta = FALSE</code> , exclude estimates of the beta matrix.
<code>nu</code>	Logical. If <code>nu = TRUE</code> , include estimates of the nu vector, if available. If <code>nu = FALSE</code> , exclude estimates of the nu vector.
<code>psi</code>	Logical. If <code>psi = TRUE</code> , include estimates of the psi matrix, if available. If <code>psi = FALSE</code> , exclude estimates of the psi matrix.
<code>theta</code>	Logical. If <code>theta = TRUE</code> , include estimates of the theta matrix, if available. If <code>theta = FALSE</code> , exclude estimates of the theta matrix.
<code>converged</code>	Logical. Only include converged cases.
<code>grad_tol</code>	Numeric scalar. Tolerance for the maximum absolute gradient if <code>converged = TRUE</code> .
<code>hess_tol</code>	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if <code>converged = TRUE</code> .
<code>vanishing_theta</code>	Logical. Test for measurement error variance going to zero if <code>converged = TRUE</code> .
<code>theta_tol</code>	Numeric. Tolerance for vanishing theta test if <code>converged</code> and <code>theta_tol</code> are <code>TRUE</code> .

...	additional arguments.
burnin	Integer indicating initial samples to discard. If burnin = NULL, do not discard anything.

Value

Returns a list of sampling variance-covariance matrices.

Returns the sampling variance-covariance matrix of the estimated parameters.

Returns the sampling variance-covariance matrix of the estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

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